Google Student Researcher Interview Study Plan

Reweighted plan starting June 9 **|** *Interview expected June 22 or later*

Project: OTOC / measurement-damage echo experiments on Willow; random circuits, scrambling, RMT, Cirq, and hardware error mitigation

1. Strategy: weighted, not replaced

This plan incorporates what has already been completed, Rodrigo's latest guidance, and the original theory requirements. The new topics (Pauli twirling, circuit folding, ZNE, Kraus/POVM, Lieb-Robinson bounds, light-cone pruning) do not replace the core project theory. They sit on top of it. The goal is to be able to explain the ideal protocol, implement and plot toy versions in Cirq/ReCirq, and reason about noisy hardware data.

Pillar	Weight	What you must be able to do
Random circuits + Haar/RMT + 1/2 vs 3/4	25%	Explain what a random circuit is, what Haar-random behavior means, why finite depth matters, and why the project expects a quantum plateau different from the classical 1/2 baseline.
OTOC / Loschmidt echo / measurement damage	25%	Explain U -> damage/butterfly -> U^dagger, distinguish Pauli butterfly from mid-circuit measurement, and model measurement as a channel.
Cirq / ReCirq implementation and plotting	20%	Build U, U^dagger, mid-circuit measurement, run trials, average random instances, and plot recovery probability or OTOC-style signals.
Error mitigation	15%	Explain coherent vs stochastic error, Pauli twirling, circuit folding, and zero-noise extrapolation (ZNE).
Light cones / square grid / butterfly velocity	10%	Use the square-grid geometry to reason about operator spreading, Lieb-Robinson bounds, butterfly velocity, and light-cone pruning.
Research fit	5%	Connect KPOs, FOTOCs, quantum chaos, RMT, leakage, and open systems to the project without overclaiming.

2. Current status as of June 9

- Completed Cirq basics from the official website.
- Built a plain-Cirq echo circuit with a mid-circuit measurement damage step.
- Ran many random trials and obtained a toy 3/4-like plateau for 6 LineQubits in simulation.
- Read the main text of Mi et al., "Information Scrambling in Computationally Complex Quantum Circuits" (not appendix yet).
- Read the Scholarpedia overview on OTOCs and quantum chaos.

Important caution: your 6-qubit LineQubit result is a valuable sanity check, but do not present it as reproducing the full Willow/RMT result. Say: "I built a toy measurement-damage echo that already shows a 3/4-like plateau, but I treat it as a preparation check. The next steps are square-grid geometry, the correct random-circuit ensemble, ReCirq/OTOC workflow, and noise mitigation."

3. Rodrigo's guidance: how each topic fits

Topic Rodrigo mentioned	How to understand it	Where it enters the plan
Measurement is similar to Pauli butterfly	A discarded Z-basis measurement is a non-unitary channel M_Z(rho)=P0 rho P0 + P1 rho P1 = 1/2(rho + Z rho Z). It is an average over I and Z insertions, not a single unitary gate.	June 11; also use when explaining why the project is OTOC-like.
Pauli twirling	Randomizes coherent, structured gate errors into an effectively stochastic Pauli error model. It does not remove error; it changes the error model.	June 15.
Coherent vs incoherent error	A coherent over-rotation can accumulate systematically over many gates. Twirling makes the accumulation more stochastic and easier to model.	June 15.
Circuit folding	Replace U by U U^dagger U (or local folds) to increase effective noise while ideally preserving the logical circuit.	June 16.
ZNE	Run the same logical observable at several noise	June 16 and NMR/Willow

	levels and extrapolate back to zero noise.	reading.
Kraus operators and POVMs	Formal language for measurements, quantum channels, and nonselective measurement damage.	June 11.
Identify gates that do not contribute	This is light-cone pruning, not generic compilation: remove gates outside the causal cone of the observable/damage/probe.	June 17.
Lieb-Robinson bounds	A speed limit for operator/information spreading in local systems; motivates butterfly velocity and light cones.	June 17.
Square grid	The project uses N=9 or N=16, so move beyond LineQubits to 3x3 and 4x4 GridQubits.	June 10.

4. Core theory subjects to keep

Subject	Minimum interview-level understanding
Random circuit	A layered circuit with randomly chosen local gates and entangling gates. It is a controllable model of generic many-body quantum dynamics. It is not literally Haar-random at finite depth.
Haar measure	The uniform probability measure over the unitary group. A Haar-random unitary is the ideal maximally random unitary. Random circuits approximate selected Haar moments after sufficient depth.
Unitary design	A finite ensemble that reproduces Haar averages up to a given moment/order. Useful because an experiment may not need full Haar randomness to reproduce the target observable.
RMT plateau	In the scrambling/random-unitary regime, selected observables become universal and independent of microscopic circuit details. The project target is the quantum 3/4 recovery probability rather than the classical 1/2 baseline.
Loschmidt echo	No-damage control: prepare state, apply U, apply U^dagger, measure. In ideal simulation recovery is 1; on hardware its decay diagnoses imperfect inversion/noise.
OTOC	A correlator such as <W^dagger(t) V^dagger W(t) V> that detects whether a time-evolved local operator has spread enough to fail to commute with another operator.
Measurement damage	A mid-circuit measurement is a channel. If the outcome is discarded, a Z-basis measurement is local dephasing and equals an average over I and Z Pauli insertions.
Finite-size/depth caveats	N=9 and N=16 are not asymptotic Haar systems. Need depth scans, random-instance averaging, controls, and noise models.

5. Daily plan from June 9 to interview

June 9 (Mon) - Rebalance + Haar/RMT/random-circuit theory

Do	• Write a clean note explaining what a random circuit is, why it is not automatically Haar-random, and what it means to approximate Haar/unitary-design behavior. • Write the safe explanation of the 3/4-like toy result: useful sanity check, not full proof. • List assumptions in your current LineQubit simulation: N=6, line geometry, simple random-circuit ensemble, ideal simulator, no hardware noise. • Start the Kraus/POVM formulas but do not go deep yet.
Read / watch	• Review Scholarpedia OTOC overview notes you already made. • Read short sections on Haar random/unitary design from Roberts & Yoshida or a trusted set of notes. • Skim the project description again and identify where random circuits, RMT, and 3/4 enter.
Deliverable	• One-page note: random circuits, Haar measure, unitary design, RMT plateau, finite-depth caveat. • One paragraph: how to explain your 6-qubit toy 3/4 result without overclaiming.

June 10 (Tue) - Square-grid Cirq: move beyond LineQubits

Do	• Replace LineQubit.range(6) with GridQubit.rect(3,3) for N=9. • Build a square-grid random circuit with alternating

	horizontal/vertical nearest-neighbor entangling layers. • Choose a probe qubit and several damage qubits at different Manhattan distances. • Plot P(probe=1) vs depth/cycles for different damage locations. • If time allows, sketch how this would extend to GridQubit.rect(4,4) for N=16.
Read / watch	• Cirq GridQubit and circuit construction docs. • Revisit Fig. 1 and Fig. 2 of the 2021 Google scrambling paper for spatial/operator-spreading intuition.
Deliverable	• Notebook: square_grid_measurement_damage_echo.ipynb. • Figure: N=9 grid, P(probe=1) vs cycles for multiple damage locations. • Short note: what changes when moving from line to square grid.

June 11 (Wed) - Kraus operators, POVMs, and measurement damage

Do	• Study Kraus operators: E(rho)=sum_k E_k rho E_k^dagger and sum_k E_k^dagger E_k=I. • Study POVM elements: M_k=E_k^dagger E_k and p_k=Tr(M_k rho). • Derive for Z-basis projective measurement: M_Z(rho)=P0 rho P0 + P1 rho P1 = 1/2(rho + Z rho Z). • Compare your mid-circuit measurement with an explicit random I/Z insertion in Cirq and check whether averaged results agree qualitatively.
Read / watch	• Brun, "A simple model of quantum trajectories" (conceptual sections only). • Cirq noisy simulation/channel documentation if you want implementation support.
Deliverable	• One-page note: measurement damage as a dephasing channel and average over Pauli insertions. • One practiced answer: "Measurement is close to a Pauli butterfly, but it is a channel."

June 12 (Thu) - Official ReCirq OTOC workflow

Do	• Run the official Cirq/ReCirq OTOC example in reduced form: few cycles, few trials. • Identify U, U^dagger, butterfly qubit, measurement qubit, ancilla, no-butterfly normalization, raw signal, and normalized OTOC. • Save or screenshot one raw plot and one normalized OTOC-style plot. • Write a table comparing your measurement-damage echo to the ReCirq Pauli-butterfly OTOC.
Read / watch	• Official Cirq/ReCirq quantum scrambling experiment example. • Mi et al. 2021 Fig. 1 caption.
Deliverable	• One ReCirq notebook/Colab copy that runs in reduced mode. • One table: toy measurement-damage echo vs ReCirq OTOC protocol. • One paragraph answer: how to run and plot an OTOC in Cirq/ReCirq.

June 13 (Sat, light) - Error mitigation overview: Rodrigo videos

Do	• Watch Rodrigo-recommended ZNE lecture I and II if possible. • Take notes only; no heavy coding. • Define coherent error, stochastic error, Pauli twirling, circuit folding, and ZNE in your own words.
Read / watch	• Recommended ZNE/mitigation video lectures I-II. • Optional: IBM or other concise documentation on ZNE stages (noise amplification + extrapolation).
Deliverable	• Half-page summary: Pauli twirling, folding, ZNE, and why these matter for U -> U^dagger echo circuits.

June 14 (Sun, light) - Rest + review

Do	• Watch lecture III only if you have energy. • Review your random-circuit, measurement-channel, and ReCirq notes aloud. • No new heavy coding.
Read / watch	• Your notes only, plus optional video lecture III.
Deliverable	• 15-minute spoken review: project protocol, measurement damage, Cirq workflow, and 3/4 caveat.

June 15 (Mon) - Pauli twirling and coherent vs stochastic errors

Do	• Study coherent over-rotation: repeated small angle error can accumulate systematically. • Study how Pauli twirling/randomized compiling changes coherent error into an effective stochastic Pauli model. • Optional code: compare repeated coherent Rx(epsilon) errors with randomized/twirled variants. • Connect twirling to OTOC/echo experiments: coherent errors in U and U^dagger can bias the recovery signal.
Read / watch	• Short references on Pauli twirling/randomized compiling. • Your notes from Rodrigo's error-mitigation videos.
Deliverable	• One-page note: coherent vs stochastic error and why Pauli twirling helps. • 60-second interview answer on Pauli twirling.

June 16 (Tue) - Circuit folding and zero-noise extrapolation (ZNE)

Do	• Implement or at least write pseudocode for global folding: U -> U U^dagger U. • Add a simple depolarizing or amplitude damping noise model to your toy echo circuit. • Measure P(probe=1) at fold/noise scale factors 1, 3, and 5 if feasible. • Fit a simple linear/quadratic extrapolation to noise scale 0. • Connect this to the NMR/Willow paper appendix/error-mitigation discussion.
Read / watch	• ZNE lecture notes/videos. • NMR/Willow paper sections or appendix discussing zero-noise extrapolation. • Cirq noisy simulation docs as needed.
Deliverable	• One plot or sketch: observable vs noise scale with zero-noise extrapolation. • One practiced answer: "What is circuit folding? What is ZNE?"

June 17 (Wed) - Lieb-Robinson bounds, butterfly velocity, and light-cone pruning

Do	• Study the basic Lieb-Robinson idea: locality imposes an effective speed limit on information/operator spreading. • Connect butterfly velocity to the OTOC/operator front. • Use your 3x3 square-grid plot to interpret damage-qubit distance and onset depth. • Write a conceptual description of light-cone pruning: remove gates outside the causal cone of the observable, not just generic compiler simplification.
Read / watch	• Mi et al. 2021 sections/figures on butterfly velocity and operator spreading. • A short conceptual reference on Lieb-Robinson bounds; no proof required. • Nahum, Vijay, Haah PRX 2018 only if you want deeper random-circuit operator-spreading language.
Deliverable	• One diagram or note: square-grid causal cone from probe to damage qubit. • 60-second answer: Lieb-Robinson bound, butterfly velocity, and light-cone pruning.

June 18 (Thu) - 2021 paper targeted deepening

Do	• Re-read the 2021 Google scrambling paper figures and

	methods/appendix only where relevant. • Focus on normalization, operator spreading, operator entanglement, butterfly velocity, and hardware error controls. • Do not read appendix blindly; extract only what supports interview questions.
Read / watch	• Mi et al. Science 2021 / arXiv 2101.08870: figures, captions, normalization, methods relevant to OTOC measurement.
Deliverable	• One-page summary: what the 2021 paper teaches for Rodrigo's project. • Table: 2021 Pauli-butterfly protocol vs Rodrigo measurement-damage protocol.

June 19 (Fri) - 2025 papers: OTOC(2), Quantum Echoes, and NMR/Willow ZNE

Do	• Read the 2025 Nature paper at high level: abstract, Fig. 1, Fig. 2, repeated time reversal, OTOC(2) as interferometer. • Read NMR/Willow paper only for context: many-body echoes, Willow simulation, fSim/swap-network/native-gate logic, ZNE/Pauli-path mitigation. • Prepare a short explanation of why these papers are relevant but not the same as the project.
Read / watch	• Nature 2025 OTOC(2)/Quantum Echoes paper. • Quantum computation of molecular geometry via many-body nuclear spin echoes, especially error-mitigation/ZNE parts.
Deliverable	• Five bullets on OTOC(2)/Quantum Echoes. • Five bullets on NMR/Willow and ZNE relevance.

June 20 (Sat, light) - Mock interview 1 and repair weak points

Do	• Run a 45-minute mock interview. • Include questions on random circuits/Haar, OTOC, measurement channel, Pauli twirling, ZNE, Lieb-Robinson, and Cirq workflow. • Identify the weakest two answers and repair them.
Read / watch	• Your one-page summaries only.
Deliverable	• List of weak points and improved answers. • One spoken pass through your 2-minute research pitch.

June 21 (Sun, light) - Final rehearsal

Do	• No new papers. No heavy debugging. • Review the minimum checklist. • Practice slow, assumption-explicit answers: "My understanding is..." and "If the outcome is discarded..." • Prepare 5-8 questions for Rodrigo/interviewer.
Read / watch	• Your notes only.
Deliverable	• Calm final checklist. • Final questions list.

June 22 or interview day - Interview mode

Do	• Review one page only: protocol, random circuits/Haar/RMT, measurement channel, Cirq workflow, error mitigation, light cone, research pitch. • Speak slowly. Do not overclaim the toy 3/4 result. • If unsure, state the assumption: e.g., "assuming the mid-circuit measurement outcome is discarded..." • Ask questions about observable definition, postselection/conditioning, random-circuit ensemble, finite-size corrections, and noise controls.
Read / watch	• No new reading.
Deliverable	• Clear research-level conversation.

6. Answers to practice until automatic

Q: What have you done so far?

A: I went through Cirq basics, built a toy measurement-damage echo in plain Cirq, averaged over random trials, and observed a 3/4-like recovery plateau for a 6-qubit LineQubit simulation. I treat this as a sanity check, not the full Willow/RMT result. I also read the main text of the 2021 Google scrambling paper and the Scholarpedia OTOC overview.

Q: What is a random circuit?

A: A random circuit is a layered circuit with randomly chosen local gates and entangling gates. It is a controllable model of generic many-body dynamics. A finite-depth random circuit is not exactly Haar-random, but it can approximate selected Haar moments or unitary-design behavior after enough depth.

Q: Why is the mid-circuit measurement similar to a Pauli butterfly?

A: If a Z-basis measurement outcome is discarded, the channel is M_Z(rho)=P0 rho P0 + P1 rho P1 = 1/2(rho + Z rho Z). So it is equivalent to averaging over no Pauli insertion and a Z insertion. That makes it closely related to a butterfly OTOC, but it is still a non-unitary measurement channel.

Q: What is Pauli twirling?

A: Pauli twirling/randomized compiling randomizes the error frame so coherent, structured gate errors become effectively stochastic Pauli-like errors. It does not remove noise, but it makes the noise easier to model and mitigate.

Q: What is circuit folding and ZNE?

A: Circuit folding replaces U by something like U U^dagger U, which is ideally the same logical operation but with more gates and therefore more noise. ZNE measures the observable at several effective noise levels and extrapolates back to the zero-noise limit.

Q: What is Lieb-Robinson / butterfly velocity?

A: The Lieb-Robinson idea is that in a local system information and operator support cannot spread infinitely fast. In OTOCs this appears as a front; the butterfly velocity is the speed of that front. It motivates light-cone pruning in circuits.

Q: How does your research connect?

A: I believe my research connects to the proposed project mainly through **QI spreading** and **echo diagnostic**. First, my current work on coupled Kerr-cat qubit focus in describing when this particular system is regular, chaotic, or mixed. We utilize a similar OTOC called the FOTOC for Info spreading, also spectral statistics, entanglement entropy, RMT to charactize when the system behaves chaotically and how Info spreads due to it.

For the project I understand **random circuit** is a different platform but the **diagnostic is very similar**. Forward evolution, local perturbation, backward evolution, and measure how much info was recovered. That is basically what FOTOC does U perturbation U^dagger and the measure the **sensitivity (Q and P)**. Also, using the random matrix bahavior to connect to quantum chaotic system is very familiar to me, in fact, I have been working on that since my undergrad when I started my research on quantum chaos.

Finally, the same coupled Kerr-cat qubit project aims to study 2Q gate fidelity under chaoticity problem (being a chaotic system does that influence leakage deu to the info spreading? How dissipation deals with it, does it help or make worse?) These are problems that I have being thinking already, and I believe they can be readly translated to the project by how to differentiate info spreading from the circuit or from noise, when noise mitiationg is needed.

Q: What is the usual main bottleneck when Simulating in Willow?

A: M What do you expect to be the main bottleneck: circuit depth, gate fidelity, mid-circuit measurement fidelity, leakage, calibration drift, or statistical averaging over random circuits?

Q: Are we also making classical simulation?

A: For $N = 9$and $N = 16$, will part of the work involve classical simulation of the same circuits for verification and finite-size analysis? Classical means probabilistic states or popolution in markovian process?

7. References to mention or cite to Rodrigo

- **Cirq/ReCirq OTOC example:** Google Quantum AI, "Quantum scrambling experiment example." Practical build/run/plot reference for the 2021 OTOC protocol. https://quantumai.google/cirq/experiments/otoc/otoc_example
- **Google 2021 scrambling experiment:** X. Mi et al., "Information scrambling in quantum circuits," Science 374, abg5029 (2021); arXiv:2101.08870. Use for random-circuit OTOC protocol, butterfly operator, normalization, operator spreading, and operator entanglement.
- **Scholarpedia OTOC overview:** Scholarpedia, "Out-of-time-order correlations and quantum chaos." Use for broad OTOC/chaos conceptual background.
- **Kraus/POVM and quantum trajectories:** T. A. Brun, "A simple model of quantum trajectories," arXiv:quant-ph/0108132. Use for measurement, monitoring, and quantum trajectories.
- **OTOC tutorial:** S. Xu and B. Swingle, "Scrambling Dynamics and Out-of-Time Ordered Correlators in Quantum Many-Body Systems: a Tutorial," arXiv:2202.07060.
- **Random circuits/operator spreading:** A. Nahum, S. Vijay, and J. Haah, "Operator Spreading in Random Unitary Circuits," Phys. Rev. X 8, 021014 (2018).
- **Unitary design / Haar connection:** D. A. Roberts and B. Yoshida, "Chaos and complexity by design," JHEP 04, 121 (2017). Use for frame potential, unitary designs, and OTOC/randomness connection.
- **NMR/Willow echo context:** C. Zhang et al., "Quantum computation of molecular geometry via many-body nuclear spin echoes," arXiv:2510.19550 (2025). Use for many-body echoes, Willow simulations, and ZNE/Pauli-path error mitigation.
- **OTOC(2)/Quantum Echoes:** Google Quantum AI and collaborators, "Observation of constructive interference at the edge of quantum ergodicity," Nature 646, 825-832 (2025). Use for repeated time reversal and higher-order OTOCs.
- **Error mitigation background:** Use Rodrigo-recommended ZNE lectures; optionally supplement with concise documentation on zero-noise extrapolation, circuit folding, and Pauli twirling/randomized compiling.

8. Minimum checklist before interview

- Can explain the project protocol in 60-90 seconds.
- Can explain random circuits, Haar measure, unitary designs, RMT plateau, and finite-depth caveats.
- Can explain Loschmidt echo, OTOC, and measurement damage as a non-unitary channel.
- Can show or verbally describe two plots: toy measurement-damage echo and ReCirq/OTOC-style plot.
- Can explain why your 6-qubit toy 3/4 result is a sanity check, not a proof of the full experiment.
- Can explain Pauli twirling, coherent vs incoherent errors, circuit folding, and ZNE.
- Can explain Lieb-Robinson bounds, butterfly velocity, and light-cone pruning on a square grid.
- Can connect KPO/FOTOC/quantum chaos/RMT/leakage work to the project in 2 minutes.
- Has 5-8 thoughtful questions ready about measurement outcome handling, finite-size corrections, random-circuit ensemble, controls, and noise mitigation.

TELL ME ABOUT YOUR BACKGROUND:

- I am currently a Physics PhD student at UConn working on nonlinear quantum systems and quantum chaos, especially superconducting-circuit-inspired systems like KPOs and Kerr-cat qubits. The central question in my research is how quantum information spreads, and how that connects to chaotic dynamics, leakage, and gate stability. To study that, I use tools like FOTOCs, spectral statistics, entanglement, and random matrix theory.
- My background is mostly theoretical and numerical. I work with Python, Julia, high-performance computing, Floquet dynamics, open-system simulations, and statistical analysis of quantum chaotic systems. I am also involved with the UConn Quantum Computing Club, where I mentor students on quantum circuits, Qiskit basics, noise models, and hardware principles. I enjoy presentations and discussions because they help me clarify ideas and learn from people with different backgrounds.
- Looking forward, I want to connect my quantum chaos background more directly to quantum hardware and quantum simulation software. This project is exciting because it uses random circuits and echo/OTOC-like protocols to study scrambling experimentally, which connects naturally to my experience with information spreading, RMT, leakage, and open-system effects.

- **UConn PhD → KPO/Kerr-cat → information spreading → FOTOC/RMT/leakage → Python/Julia/HPC → QC club → hardware + quantum simulation → Google OTOC project.**

QUESTIONS

- **Observable definition**
- Should I think of the main measured quantity primarily as a probe recovery probability, as an OTOC-like observable, or as a quantity that can be mapped to an OTOC after averaging?
- **First-month contribution**
- If I joined the project, what would be the most useful first contribution in the first few weeks: reproducing toy simulations, building Cirq workflows, analyzing data, implementing mitigation, or studying the RMT prediction more carefully?
- **Project ownership**
- Rodrigo mentioned that this role may require taking real ownership of a substantial part of the project. What would good project ownership look like in practice for a student researcher in this group?
- **Working with a busy team**
- If I am driving the project and need input from people who are busy with their own work, what is the best way to request help efficiently? For example, do people prefer short written summaries, specific technical questions, code/data snippets, or regular check-ins?
- **Independence versus check-ins**
- How independent should I be in choosing simulations, controls, and analysis directions, versus checking frequently with the team to make sure I am aligned with the project goals?
- **Examples of success**
- Are there examples of previous student researchers who succeeded in similar projects? What did they do particularly well?